Analyzing the Nonlocality of Sparse Auto-encoders

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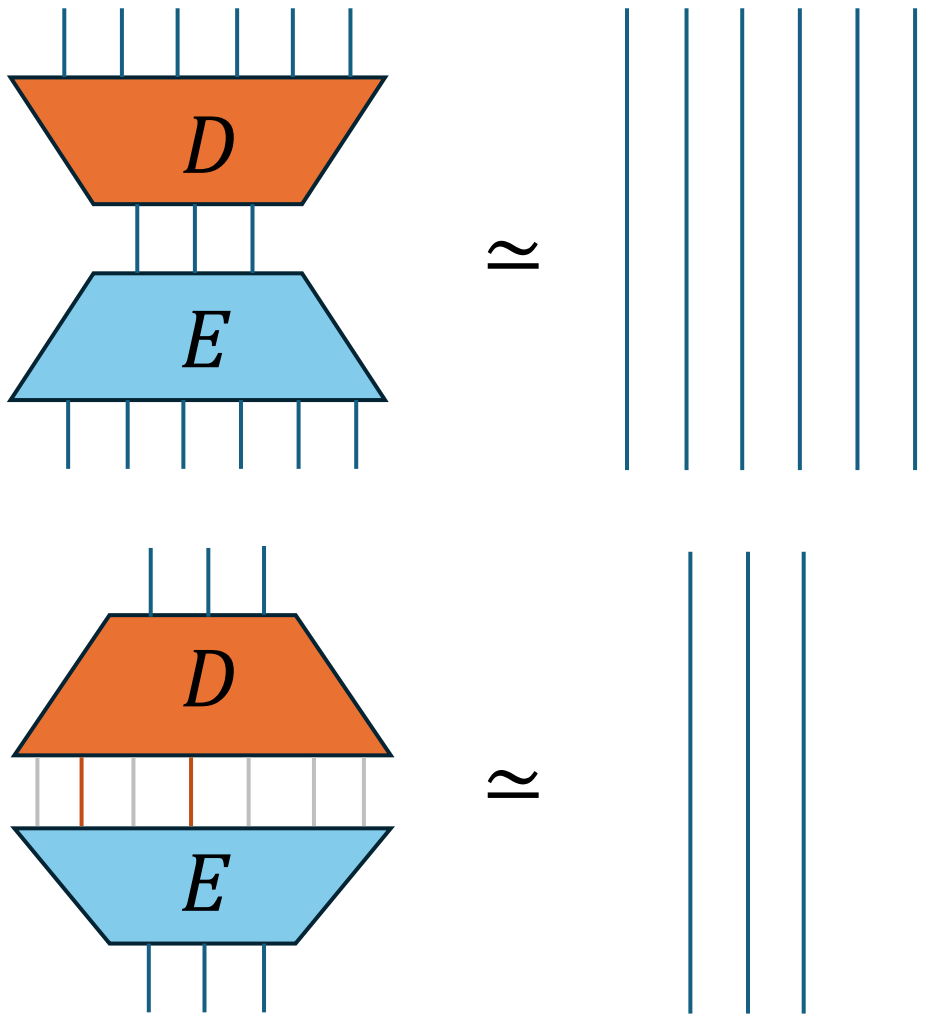
Outline

- Overview of SAE. Motivation and detailed definition
- Comparison with holographic duality.
- Analyzing the nonlocality of features
- Definition of the entropy measure
- Conclusion and discussion

Auto-Encoder and Sparse Auto-Encoder

- Feature extraction
- Ordinary auto-encoder: reproduce identity with lower dimensional representation
- Sparse auto-encoder: reproduce identity with higher dimensional representation but with sparsity condition
- Actually still a compression



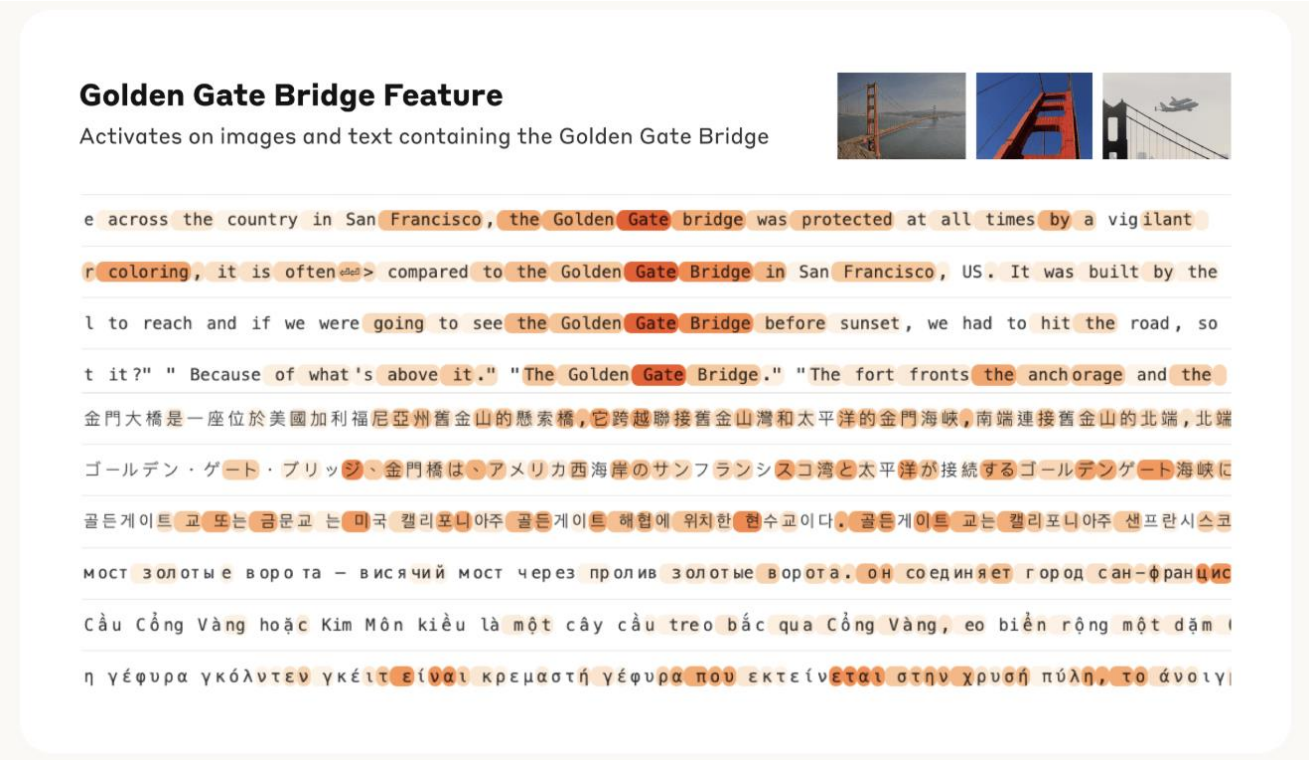
Ranzato, Boureau, LeCun '07, Ng, lecture note '11

Sparse Auto-Encoder

- Encoder $f = E(y)$ a nonlinear map
- Decoder $\tilde{y} = Df$ a linear map
- $\dim(f) > \dim(y) = \dim(\tilde{y})$
- $\tilde{y} \simeq y$
- Trained for an ensemble of y
- Probability $P(y)$ induces probability $Q(f(y)) \simeq \frac{P(y)}{|df/dy|}$ which is mainly supported on sparse vectors.

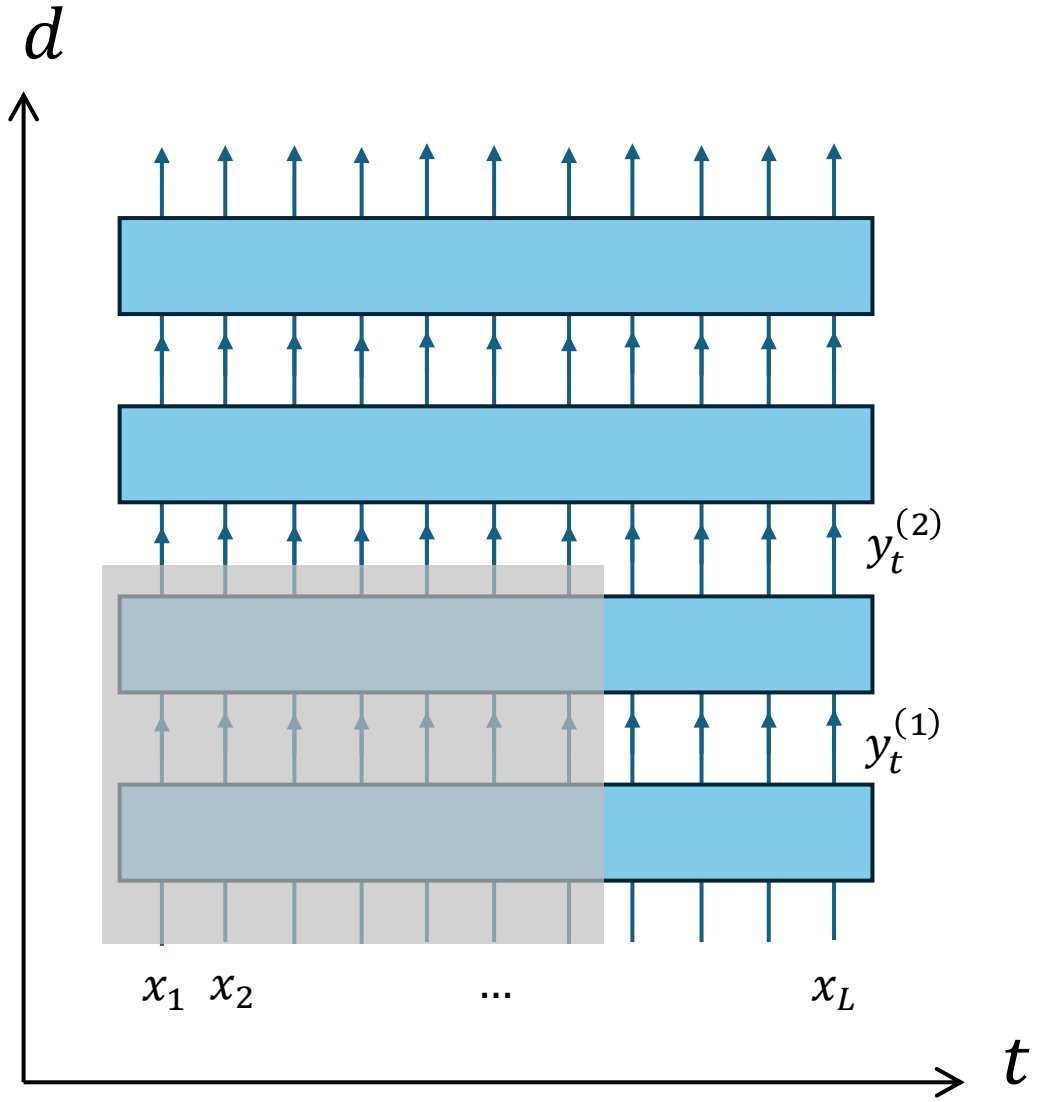
Sparse Auto-Encoder in LLM

- SAE is introduced to interpret the “mind” of LLM (Yun et al, arXiv:2103.15949)
- Example: series of research from Anthropic (https://www.anthropic.com/research/team/interpretability)



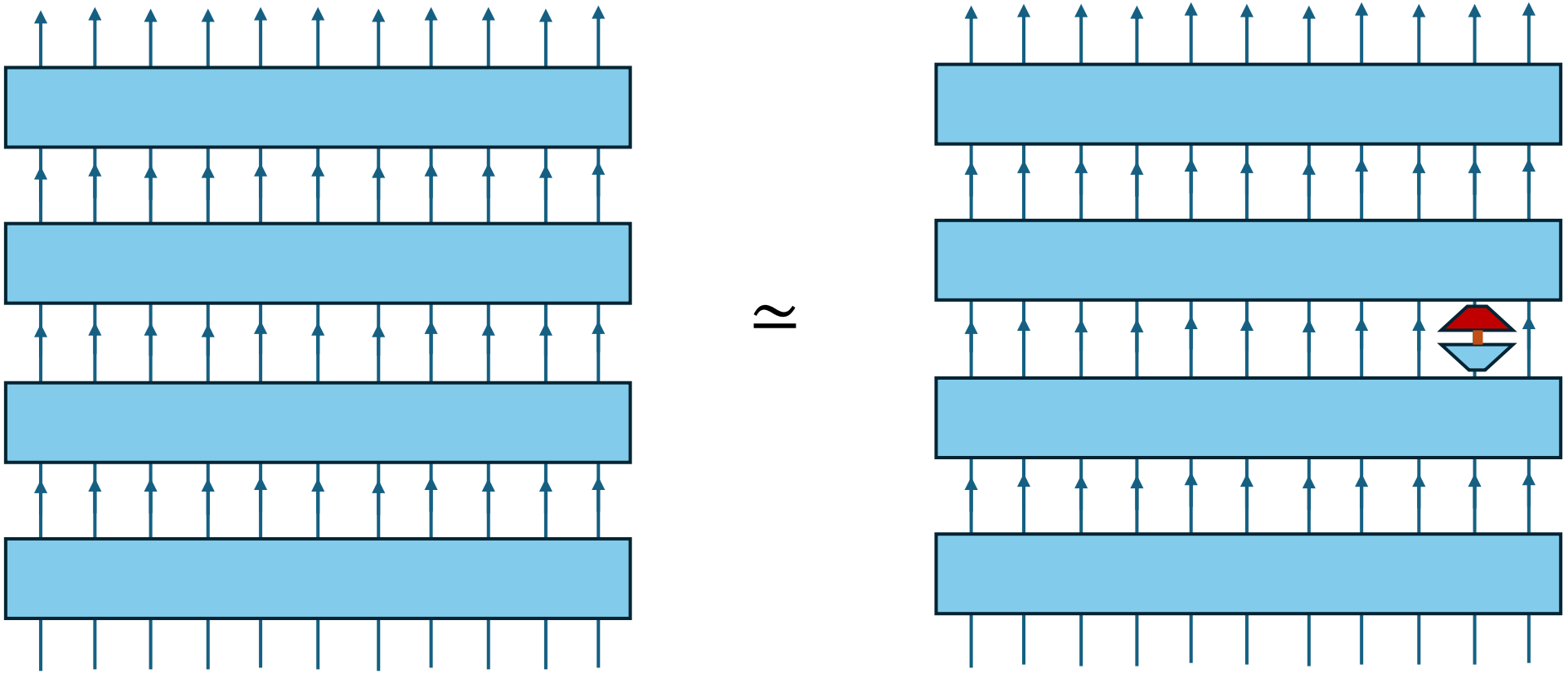
Sparse Auto-Encoder in LLM

- Causal transformer model
- Attention layer $y_t^{(n+1)} = \mathrm{Att_n}\left(y_{t'\leq t}^{(n)}\right)$
- Input token vectors $x_t = y_t^{(0)}$
- Causal structure in two dimensions



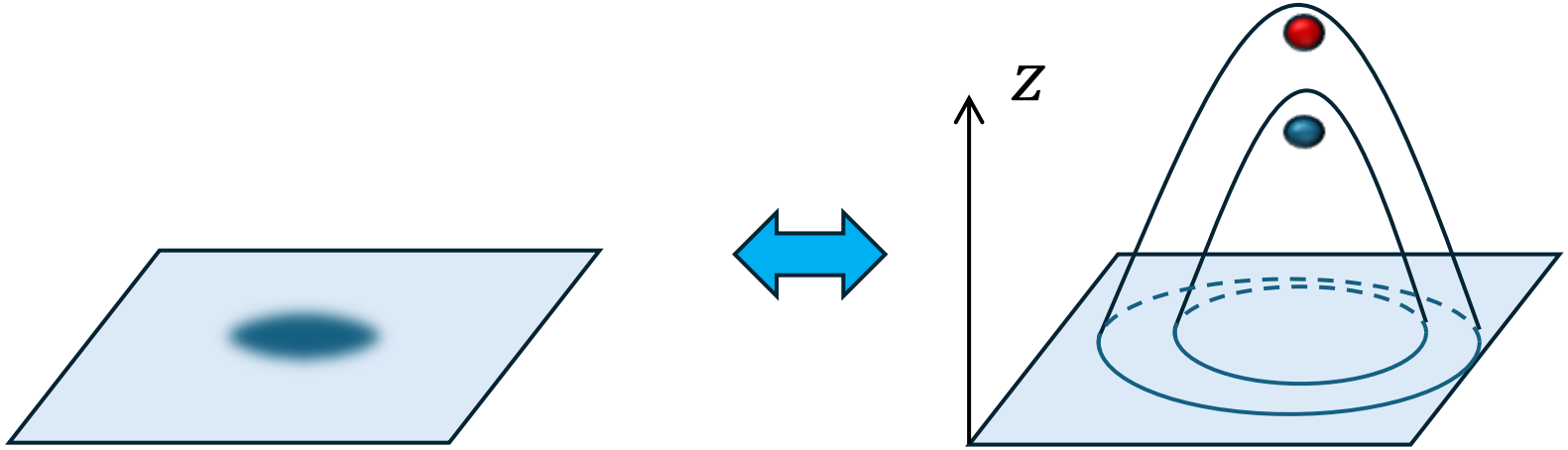
Sparse Auto-Encoder in LLM

- SAE is trained for internal layers
- High dimensional feature vector $f_t^{(n)} = E_n\left(y_t^{(n)}\right)$



SAE and holographic duality

- Ideally, SAE features represent independent concepts. For example, …
- Mapping to feature space uncovers the mind activity of LLM.
- Physics analog: bulk fields in holographic duality (Maldacena '97)

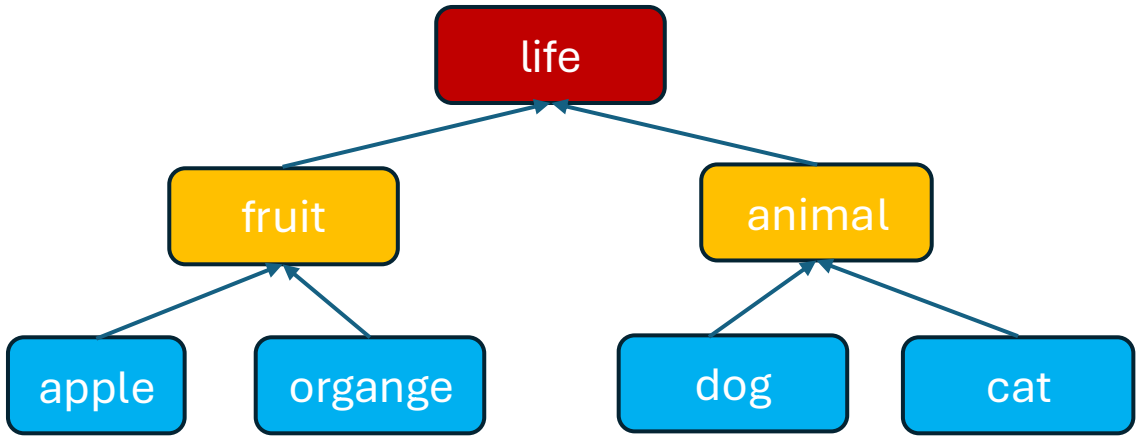


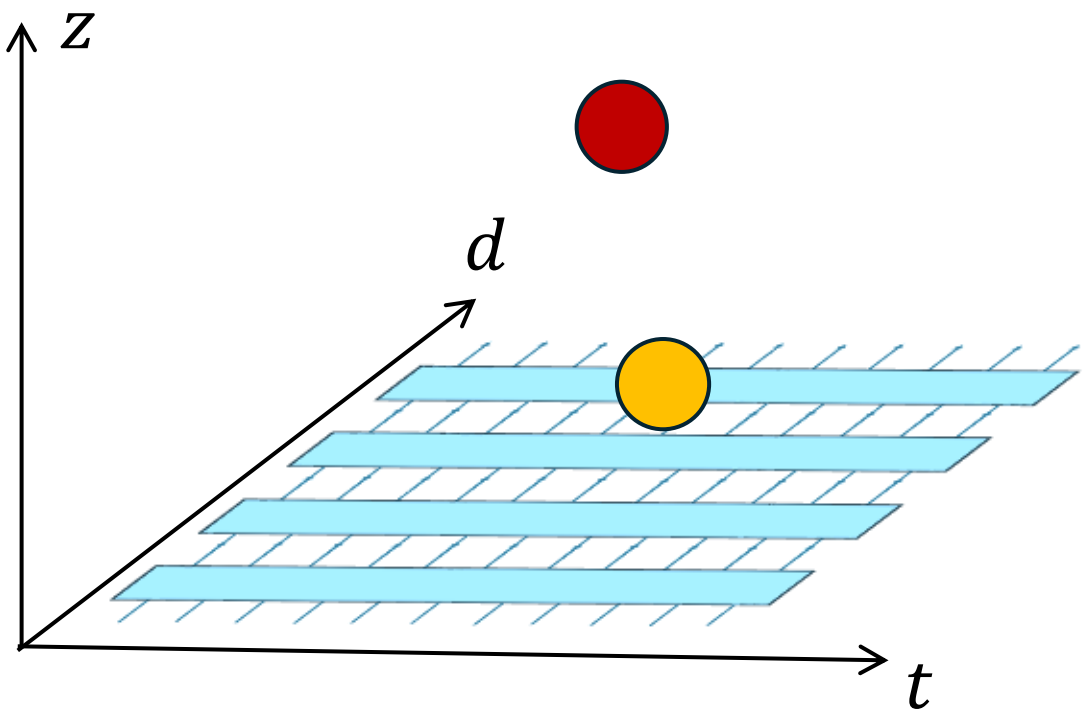
- Strongly correlated dynamics are mapped to weakly correlated "nearly free fields" in semiclassical bulk geometry

Acknowledgement: private communication with Aditya Cowsik, Kfir Dolev, Bruno de Luca

Concepts as nonlocal objects

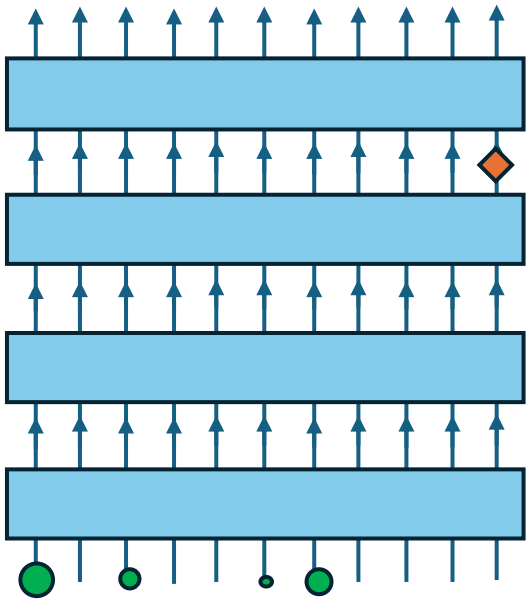
- Question: Do the SAE concepts live in an emergent new dimension?
- Inspired by holographic duality, we expect different concepts to have different nonlocality
- Different concepts correspond to different level of abstraction.





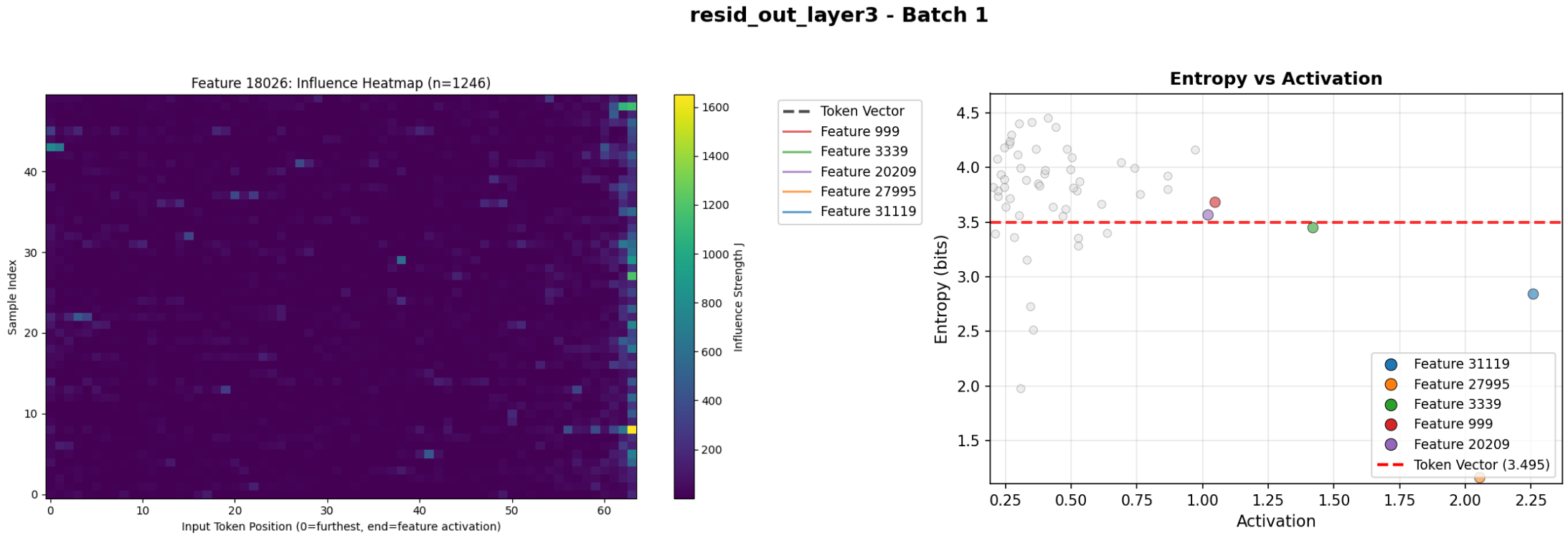
Measure the concept nonlocality

- We can measure the nonlocality of a concept by how many input tokens have significant influence to it.
- $J_a(t,n|t') = \sum_\mu I_{a,\mu}(t,n|t')^2$
- $J_a(t,n|t')$ measures the influence of token t' to concept a at location t, layer n.
- Nonlocality of the influence can be measured by entropy of the normalized influence (as a probability distribution)
- $p(t') = J_a(t,n|t')/\sum_{t'} J_a(t,n|t')$
- $S_a(t,n) = -\sum_{t'} p(t') \log p(t')$,
- Other measures such as $\sum_{t'} p(t')^2$ also possible.



Numerical results

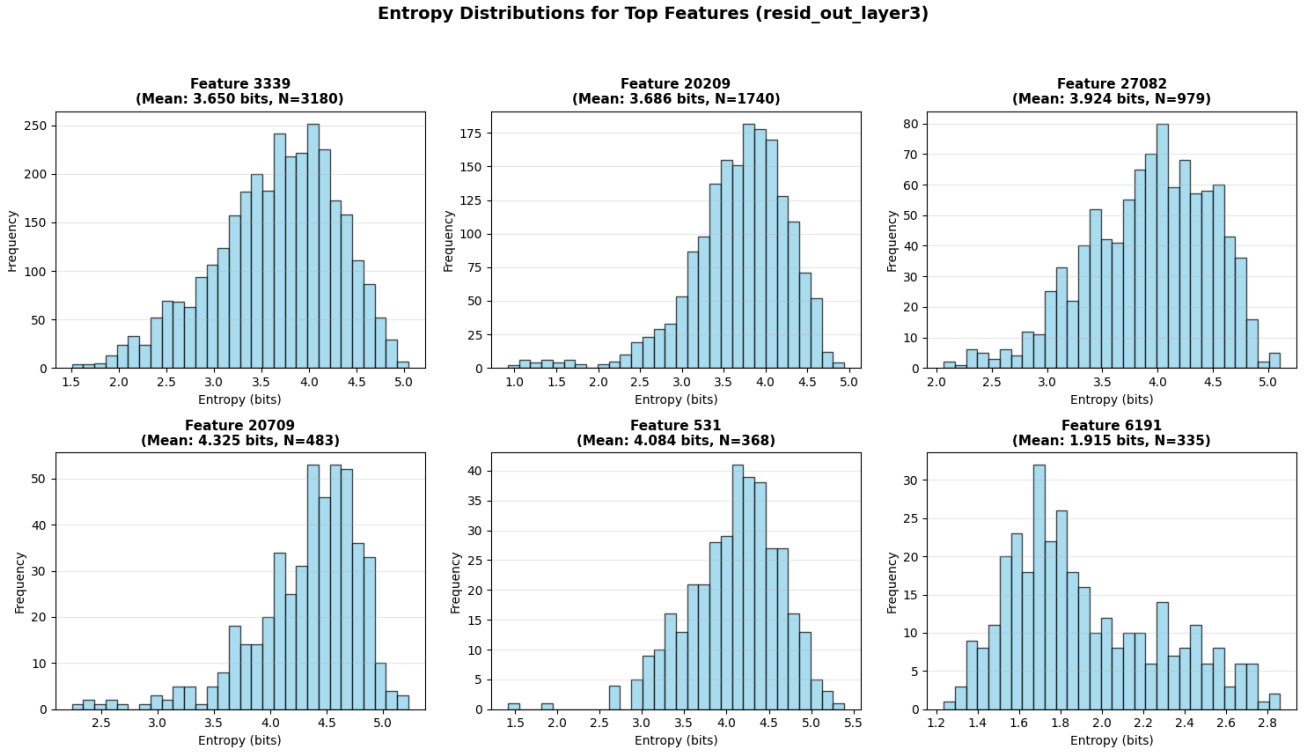
- $p(t') = J_a(t,n|t')/\sum_{t'} J_a(t,n|t')$ and the entropy $S_a(t,n)$ is computed for each batch of input token.



Numerics on a small model pythia-70m-deduped
Pretrained SAE from Marks, Karvonen, Mueller 2024 (dictionary_learning)

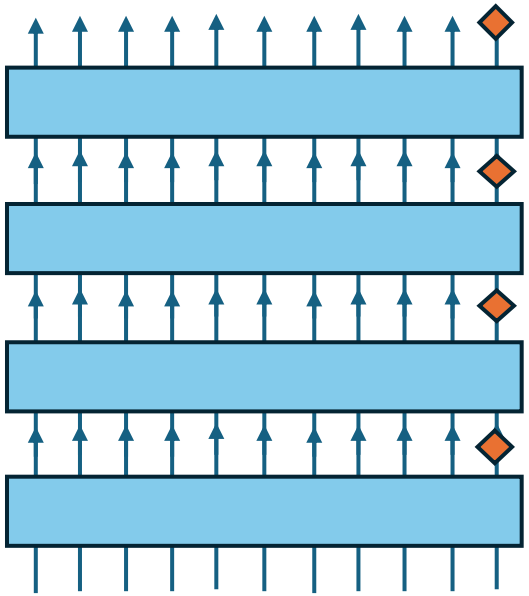
Entropy distribution across batches

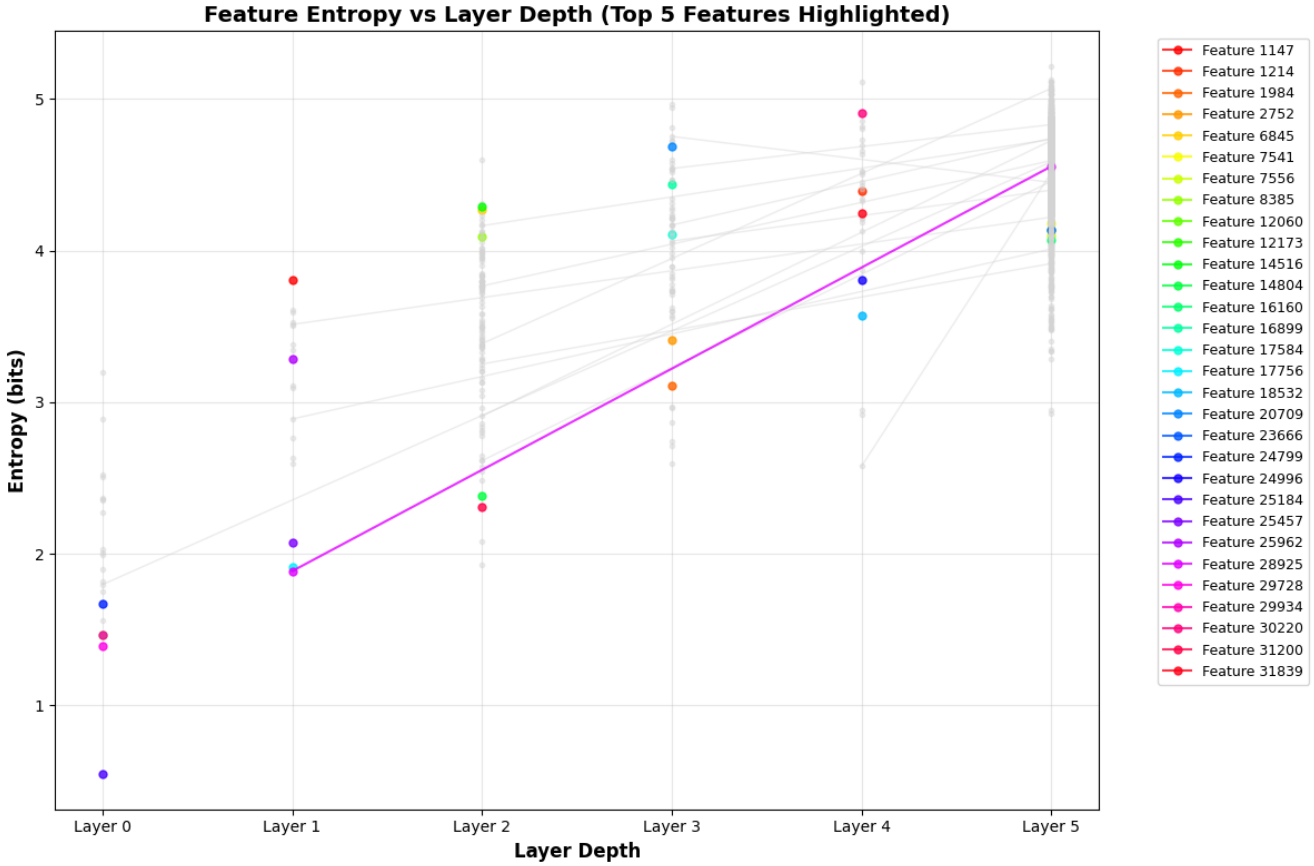
- Distribution of entropy across different batches



Depth dependence of entropy

- Deeper layers correspond to more nonlocal features
- Different layers focus on different features

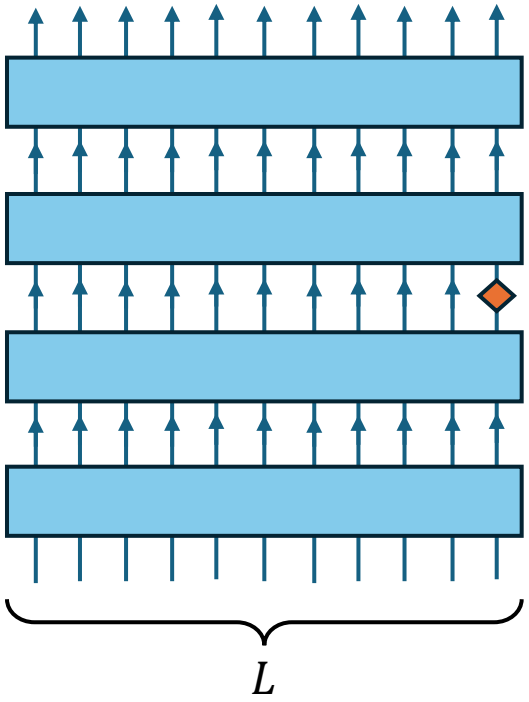


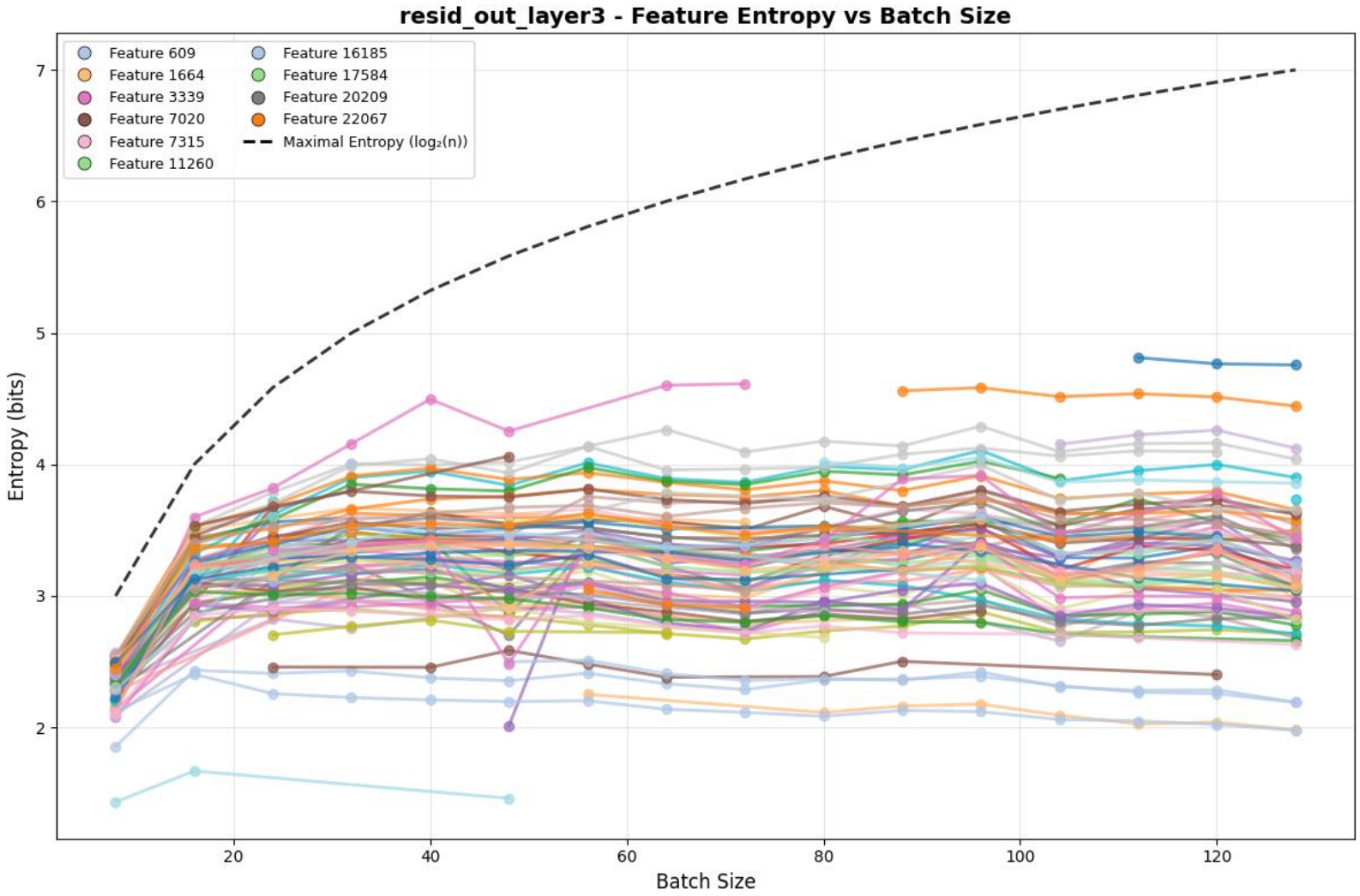


(batch size 64, for an example batch)

Length dependence of entropy

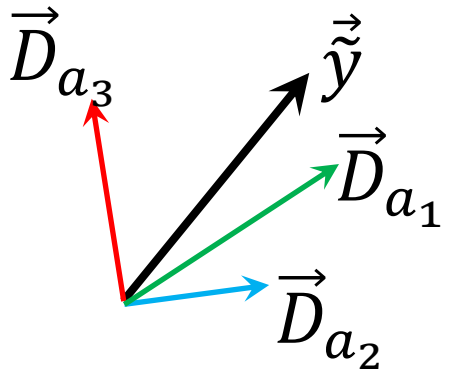
- Saturation of entropy



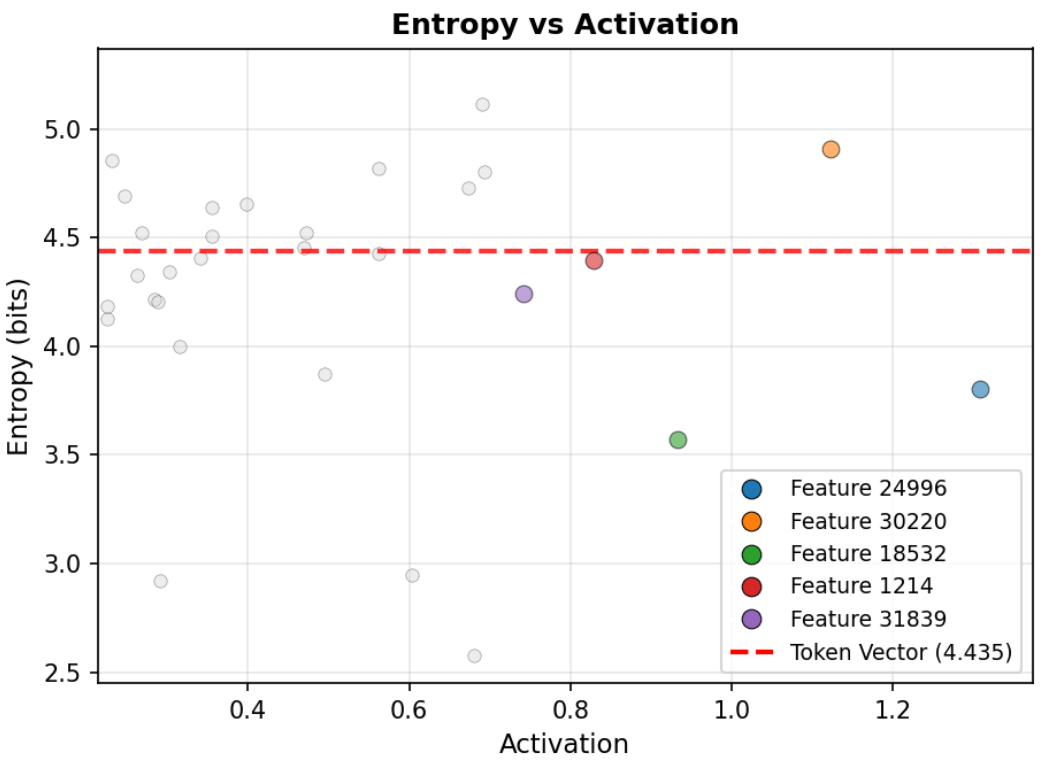


Internal layer token vector versus concepts

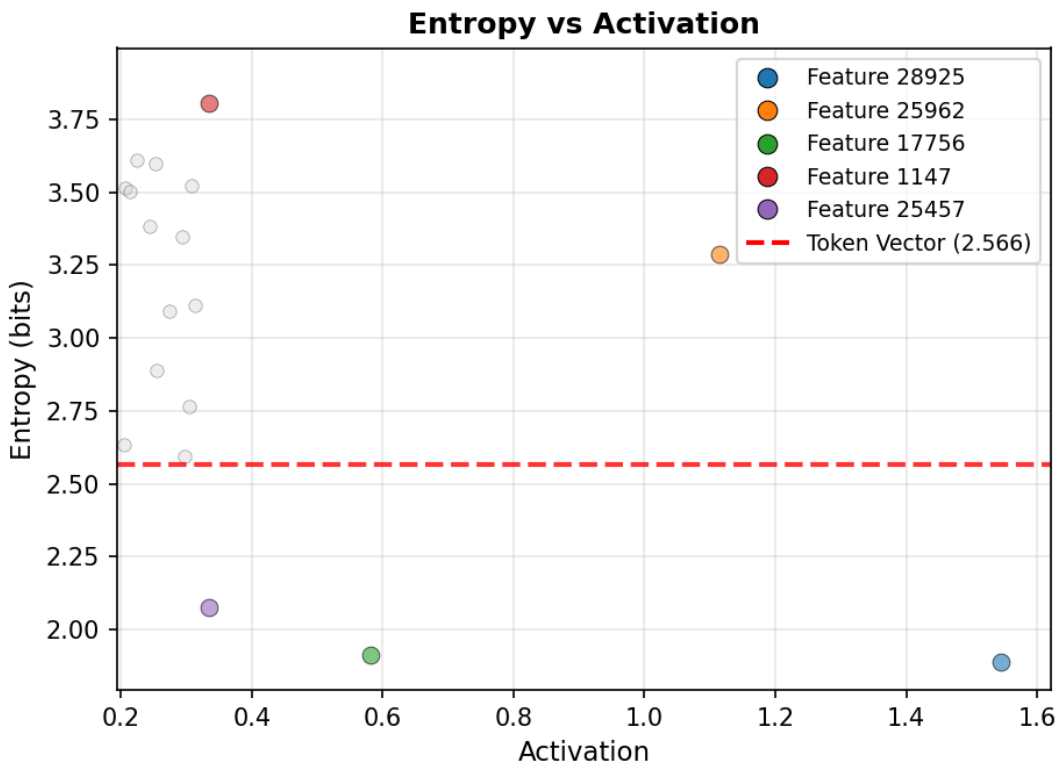
- $\tilde{y}_\mu = \sum_a D_{\mu a} f_a = D_{\mu a_1} f_{a_1} + D_{\mu a_2} f_{a_2} + \cdots$
- $\frac{d\tilde{y}_\mu(t,n)}{dx_\nu(t')} = D_{\mu a} \frac{df_a(t,n)}{dx_\nu(t')}$
- $R(t,n|t') = \left\| \frac{d\tilde{y}_\mu(t,n)}{dx_\nu(t')} \right\|_2^2$
- $q(t') = R(t,n|t') / \sum_{t'} R(t,n|t')$
- We can compare the nonlocality of the influence to embedding vector $\tilde{y}_\mu$ with that of the leading features f_a. We expect that the latter has less nonlocality in input token space, since $\tilde{y}$ has to contain information about multiple concepts that are relevant for later layers.
- Numerically, leading features tend to have lower entropy but not very consistently.



Numerical results



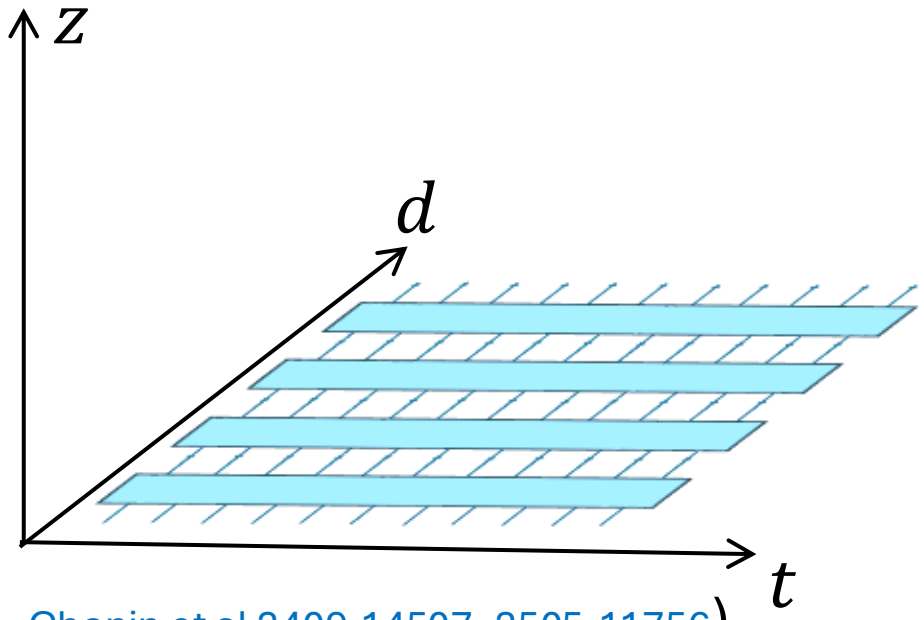
Layer depth 4



Layer depth 1

Summary and discussion

- Sparse Auto-encoder features can be characterized by the nonlocality in token space
- Nonlocality has depth dependence and length dependence
- Entropy measure
- **Open questions**:
- Other measures of nonlocality
- Criteria for good feature extraction
- Any universal behavior across models?
- Dynamical nonlocality in the token and layer dimensions (defining SAE for the union of several layers) (see e.g. Shi et al 2503.08200)
- How to learn logical relation between different level of concepts (potential problems c.f. Chanin et al 2409.14507, 2505.11756)
- General direction: Analyzing LLM from the angle of information and geometry



Collaborators

- Antigravity, Cursor and Lucien, powered by leading LLM's.

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